

DOSM Datathon 2026

“Sustainable Tourism Intelligence and Destination Capacity Monitoring System for Malaysia”

by

AYAMME

1. Nur Balqish binti Jaafar
2. Khadijah Sofia binti Anuar
3. Nadhrah Nadhirah binti Khairuddin
4. Puteri Ayu Natasha binti Megat Mohamad Hisam

*Faculty of Computer and Mathematical Sciences, Universiti Teknologi MARA, Jalan Ilmu
1/1, 40450 Shah Alam, Selangor, Malaysia*

Under the Guidance of

Dr Zainura binti Idrus

Faculty of Computer and Mathematical Sciences

Universiti Teknologi MARA (UiTM)

Date of Submission

22 September 2026

CONTENTS	PAGE
Front Page	1
Table of Contents	2
Introduction	3
Background of the Study	3
Problem Statement	3
Objectives	4
Literature Review	5
Methodology	6
Findings	8
Output (Dashboard)	11
Conclusion and Recommendations	14
Limitations	15
References	16

INTRODUCTION

Background of the Study

Tourism is an important component of Malaysia's economic and social development because it stimulates expenditure, supports employment and creates demand for accommodation, transport, food, retail and other tourism-related services. Domestic tourism is particularly important because it reflects mobility and spending within the country and can help distribute tourism benefits across states and local destinations.

The Department of Statistics Malaysia (DOSM) reported strong recovery in domestic tourism after the disruption of the COVID-19 period (DOSM, 2025). The dataset used in this study shows that domestic visitor movements continued to expand through 2025, reaching approximately 290.1 million visitor movements. The same 2025 national indicators show year-on-year growth of about 11.5%, tourism expenditure of approximately RM121.3 billion and an average length of stay of 2.56 days.

However, tourism activity is not distributed evenly across states. Some states record high visitor volumes but only moderate growth, while others are smaller markets with rapid year-on-year expansion. Visitor flows also differ in terms of where tourists originate, how strongly a destination depends on its own residents and how diversified its source markets are. These differences mean that a single tourism strategy is unlikely to be equally suitable for every state.

This study therefore applies exploratory data analysis, derived tourism-market indicators and K-Means clustering to build a state-level tourism intelligence framework. The approach supports sustainable tourism planning in the sense of better targeting promotion, managing demand, reducing over-reliance on narrow source markets and identifying different strategic needs across Malaysian states. The analysis does not claim to directly measure environmental sustainability; instead, it provides decision-support indicators for more targeted and resilient tourism planning.

Problem Statement

Malaysia's domestic tourism sector is recovering strongly, but headline visitor totals alone do not explain how tourism demand differs across states. Decision-makers need to know not only which states receive the most visitors, but also which markets are growing, which destinations rely heavily on local residents, and which destinations attract visitors from a broad range of origin states.

A second challenge is that state comparisons can be misleading if only raw visitor volume is used. A high visitor count may reflect the size of a state's population or established market position rather than

tourism intensity. Likewise, two states with similar visitor totals may have very different growth rates and source-market structures. These differences are important for planning destination promotion, infrastructure capacity and market diversification.

A third challenge is translating descriptive analysis into actionable segments. Tourism stakeholders may benefit from grouping states with similar characteristics rather than applying uniform strategies. Machine learning clustering can support this task by identifying groups of states with similar tourism profiles across multiple indicators simultaneously.

Objectives

1. To analyse domestic tourism trends and state-level visitor patterns in Malaysia from 2018 to 2025.
2. To analyse interstate tourism flows and identify destination-market characteristics using local dependency and source-market diversification indicators.
3. To segment Malaysian states using K-Means clustering based on visitor volume, year-on-year growth, local dependency, source diversification and tourism intensity.
4. To propose data-driven strategies for different tourism market segments to support more targeted and resilient tourism planning in Malaysia.

LITERATURE REVIEW

Sustainable and Data-Driven Tourism Planning

Sustainable tourism planning involves balancing tourism growth with the long-term needs of destinations, communities and tourism businesses. The United Nations Sustainable Development Goals (United Nations, n.d.) recognise tourism under Target 8.9, which calls for policies that promote sustainable tourism, and Target 12.b, which highlights the importance of tools for monitoring tourism's contribution to sustainable development. Data-driven tourism planning can support these goals by helping decision-makers understand where tourism activity is concentrated, how markets are changing and which destinations may require different strategies.

Domestic Tourism and Destination Market Structure

Domestic tourism data can reveal more than visitor totals. Origin-destination flows help identify where residents travel and how destinations depend on different source markets. A destination that relies heavily on visitors from the same state may be more exposed to local demand conditions, while a destination that attracts visitors from many origin states may have a more diversified market base. Market diversification is therefore useful as a resilience-oriented indicator, although it should not be interpreted as a complete measure of sustainability.

Machine Learning for Tourism Segmentation

Machine learning can support tourism analysis by identifying patterns across multiple variables. K-Means clustering (MacQueen, 1967) is a widely used unsupervised learning method that partitions observations into groups based on similarity in feature space. In tourism analytics, clustering can be used to identify destination profiles that are not obvious from a single indicator. Standardisation is important because K-Means is distance-based and variables measured on larger numerical scales can otherwise dominate the model. Cluster quality can be assessed using measures such as the silhouette score (Rousseeuw, 1987), while interpretation should also consider whether the resulting segments are meaningful and actionable for the intended decision context.

METHODOLOGY

Data Sources and Scope

The analysis uses four primary data components prepared in Power BI and Python. The main state-trend dataset contains domestic visitor movements by state from 2018 to 2025 (DOSM, 2025). The tourism-flow dataset contains origin-state to destination-state domestic tourism movements for 2025 (DOSM, 2025). National KPI data provide tourism expenditure and average length of stay (DOSM, 2025), while the DOSM state population dataset (DOSM, 2026) provides 2025 population values for tourism-intensity calculations.

Dataset	Key fields	Period	Purpose
State Trend	State, Year, Domestic Visitors, YoY Growth	2018–2025	Trend analysis, state ranking and growth comparison
Tourism Flow	Origin State, Destination State, Tourists	2025	Interstate flow and market-structure analysis
National KPI	Metric, Value, Year	2025	National expenditure and average length of stay
Population by State	State, Population	2025	Tourism intensity by state

Table 1. Data components used in the analysis.

Data Preparation

- Filtered the analysis to the required years and 16 Malaysian states/Federal Territories.
- Checked missing values and removed blank destination categories from final visualisations.
- Standardised field names and data types for state, year and numeric variables.
- Created a state-level machine-learning feature table for 2025.

Derived Indicators

Three derived indicators were used to move beyond raw visitor counts. Local Dependency measures the share of tourists to a destination who originate from the same state. Source Market Diversification is derived from the Herfindahl-Hirschman Index (HHI), where a higher score indicates a more diversified mix of origin states. Tourism Intensity compares visitor movements with state population.

$$\text{Local Dependency (\%)} = \text{Local tourists to destination} / \text{Total tourists to destination} \times 100$$

$$\text{Source Diversification Score} = (1 - \sum \text{source-market share}^2) \times 100$$

$$\text{Tourism Intensity} = \text{Domestic visitors ('000)} / \text{Population ('000)}$$

Exploratory Data Analysis

EDA was conducted in Power BI using national time trends, state rankings, visitor-volume versus growth scatter plots, origin-destination rankings, local-dependency comparisons and source-diversification comparisons. Interactive slicers were used only where filtering was analytically appropriate; market-structure indicators were kept independent from the origin-state slicer to avoid distorting their denominators.

K-Means Clustering

Five 2025 features were selected for clustering: Domestic Visitors, YoY Growth, Local Dependency, Source Diversification Score and Tourism Intensity. Population was not used as a separate clustering feature because it was already incorporated into Tourism Intensity. The five variables were standardised using StandardScaler before K-Means clustering (MacQueen, 1967).

Candidate values of K from 2 to 6 were evaluated. Silhouette scores (Rousseeuw, 1987) were 0.451 for K=2, 0.367 for K=3, 0.366 for K=4, 0.223 for K=5 and 0.204 for K=6. Although K=2 produced the highest silhouette score, it separated only Sabah and Sarawak from all other states and was considered too coarse for strategic use. K=4 produced a near-identical silhouette score to K=3 but created a singleton cluster containing only W.P. Kuala Lumpur. Therefore, K=3 was selected as a more interpretable and actionable segmentation. Principal Component Analysis (PCA) was used only to visualise the clusters in two dimensions and was not used to generate the cluster assignments.

FINDINGS

National Domestic Tourism Performance

Domestic tourism recovered strongly after the 2020–2021 decline. The 2025 dataset records approximately 290.1 million domestic visitor movements, representing about 11.5% year-on-year growth. Tourism expenditure reached approximately RM121.3 billion, while the average length of stay was 2.56 days. The national time series shows a clear trough in 2021 followed by sustained recovery through 2025.

State-Level Visitor Performance

Selangor recorded the highest domestic visitor volume in 2025 at approximately 36.38 million visitor movements, followed closely by W.P. Kuala Lumpur at approximately 35.06 million. Perak and Pahang also recorded high visitor volumes at approximately 23.64 million and 23.16 million respectively. The visitor-volume versus growth analysis shows that market size and growth do not always move together. W.P. Kuala Lumpur combines high visitor volume with strong growth, while Selangor has the largest visitor base but more moderate growth.

Several smaller markets recorded high growth rates. W.P. Labuan recorded approximately 34.41% year-on-year growth, W.P. Kuala Lumpur 29.93%, W.P. Putrajaya 23.06% and Perlis 16.44%. These differences indicate that growth opportunities are not limited to the states with the largest visitor volumes.

Tourism Flow and Market Structure

The origin-destination analysis shows that domestic tourists travel across state boundaries in highly uneven patterns. The dashboard allows users to select an origin state and examine its most important destination states. For example, when Selangor is selected, Perak, Melaka and Pahang are among the leading destinations in the 2025 flow data.

Local Dependency highlights the extent to which destinations rely on tourists originating from the same state. Sabah recorded the highest local dependency at approximately 75.22%, followed by Sarawak at 65.22%. Johor recorded approximately 34.84%, while Pahang and Selangor were around 23.54% and 22.92% respectively. In contrast, W.P. Putrajaya, W.P. Labuan, Perlis and Melaka recorded very low local dependency.

Source Market Diversification provides the complementary view. W.P. Putrajaya, W.P. Kuala Lumpur, Terengganu, Kelantan and Selangor recorded high diversification scores, while Sabah and

Sarawak recorded substantially lower scores at approximately 42.53 and 55.76. The combination of high local dependency and lower diversification suggests a more concentrated domestic tourism market structure in Sabah and Sarawak.

Tourism Intensity

Tourism Intensity compares visitor movements with population size. W.P. Putrajaya recorded the highest intensity at approximately 26.07 visitor movements per resident, followed by Melaka at 19.80 and W.P. Kuala Lumpur at 16.89. The indicator should be interpreted as visitor movements relative to population, not as the number of unique tourists per resident, because repeated trips may occur within a year.

K-Means State Segmentation

The K-Means model identified three distinct tourism market profiles. Cluster 0, labelled Established & Diversified Tourism Markets, contains Johor, Kedah, Kelantan, Melaka, Negeri Sembilan, Pahang, Perak, Pulau Pinang, Selangor, Terengganu and W.P. Kuala Lumpur. This cluster has an average visitor volume of about 21.59 million, average YoY growth of 10.70%, local dependency of 14.69%, diversification score of 84.36 and tourism intensity of 10.91.

Cluster 1, labelled Locally Dependent Tourism Markets, contains Sabah and Sarawak. It has average local dependency of 70.22% and a much lower average diversification score of 49.14. Cluster 2, labelled Emerging High-Growth Markets, contains Perlis, W.P. Labuan and W.P. Putrajaya. This group has relatively low visitor volume but high average growth of 24.64%, very low local dependency of 1.03%, high diversification of 86.00 and relatively high tourism intensity of 14.90.

Cluster	States	Profile	Recommended Strategy
Established & Diversified	Johor, Kedah, Kelantan, Melaka, Negeri Sembilan, Pahang, Perak, Pulau Pinang, Selangor, Terengganu, W.P. Kuala Lumpur	High visitor base, moderate growth, low local dependency, strong diversification	Sustain & Optimize
Locally Dependent	Sabah, Sarawak	Very high local dependency and lower source diversification	Diversify Source Markets

Cluster	States	Profile	Recommended Strategy
Emerging High-Growth	Perlis, W.P. Labuan, W.P. Putrajaya	Low visitor volume but rapid growth, low local dependency and high diversification	Promote & Develop Capacity

Table 2. K-Means tourism market segments and recommended strategies.

OUTPUT (DASHBOARD)

The final Power BI solution is organised into five pages. The first two pages are exploratory analysis pages used to document state-trend and tourism-flow findings. The three presentation-oriented pages are Tourism Overview, Tourism Flow & Market Structure, and ML Segmentation & Strategy.

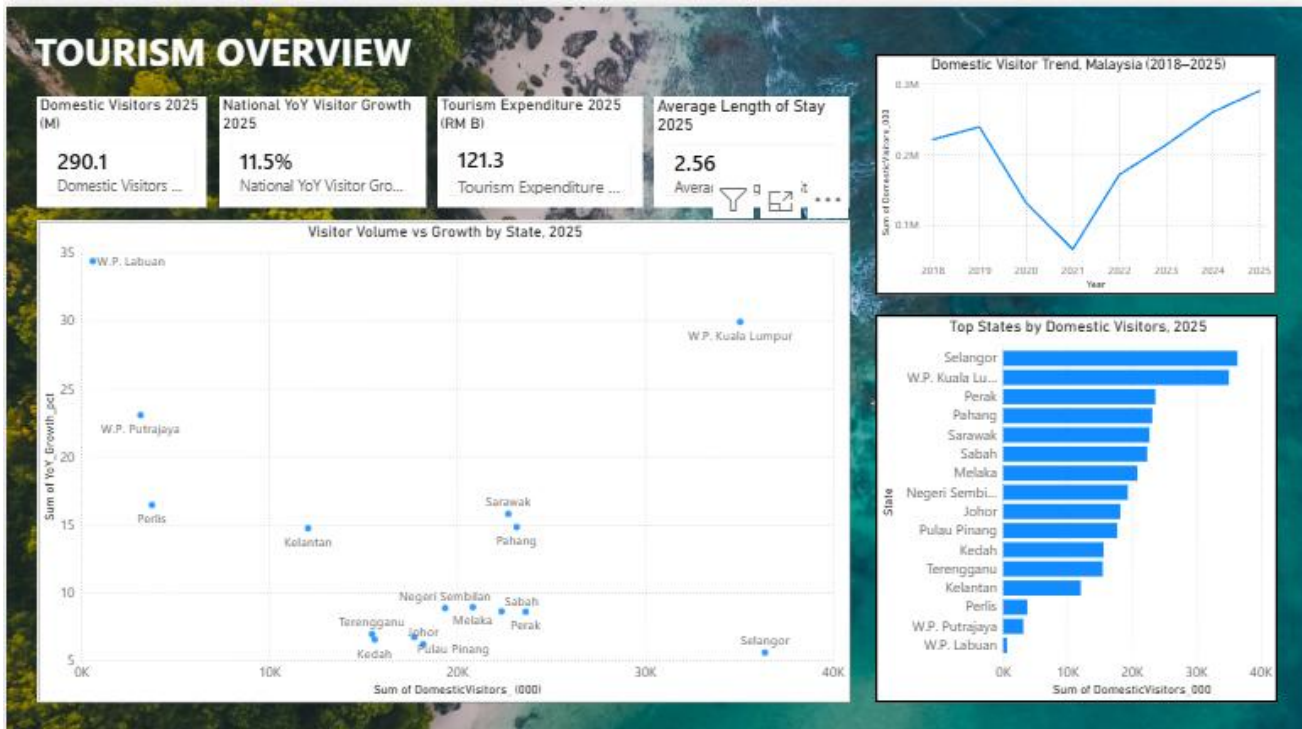


Figure 1. Tourism Overview dashboard showing national KPIs, time trend, state ranking and visitor-volume versus growth analysis.

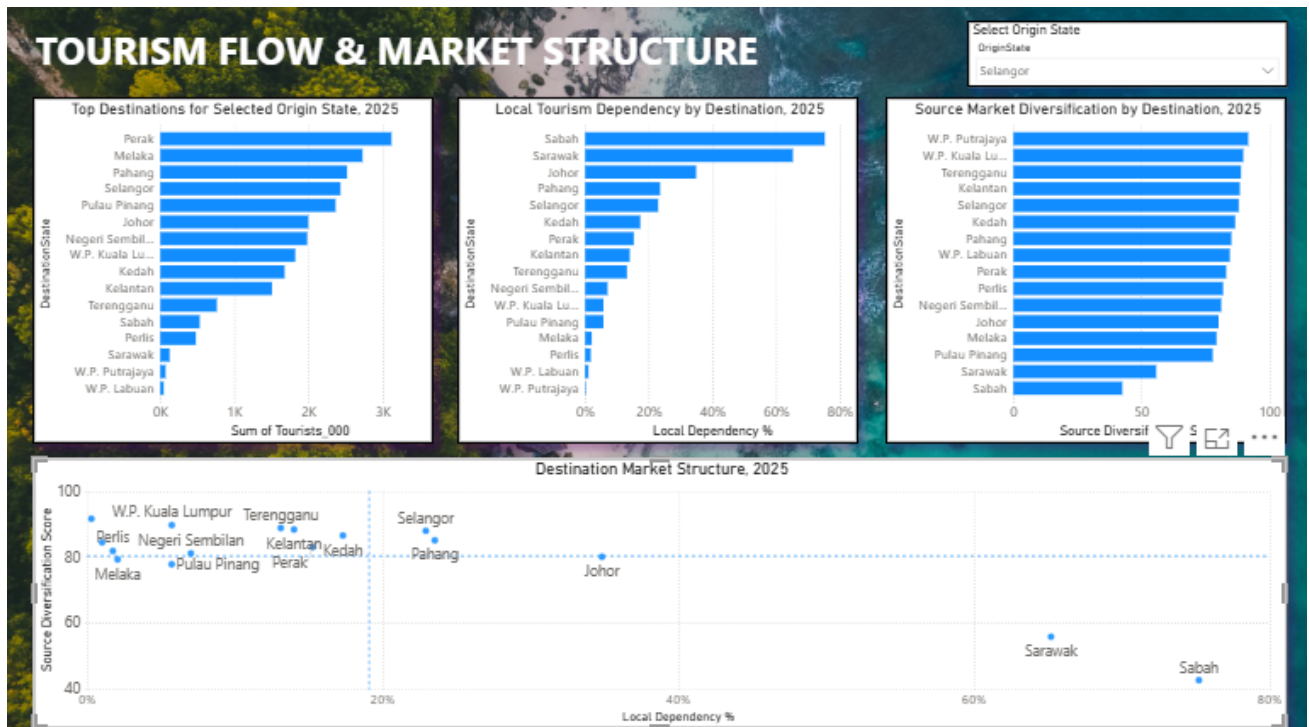


Figure 2. Tourism Flow & Market Structure dashboard with origin-state flow analysis and destination market-structure indicators.

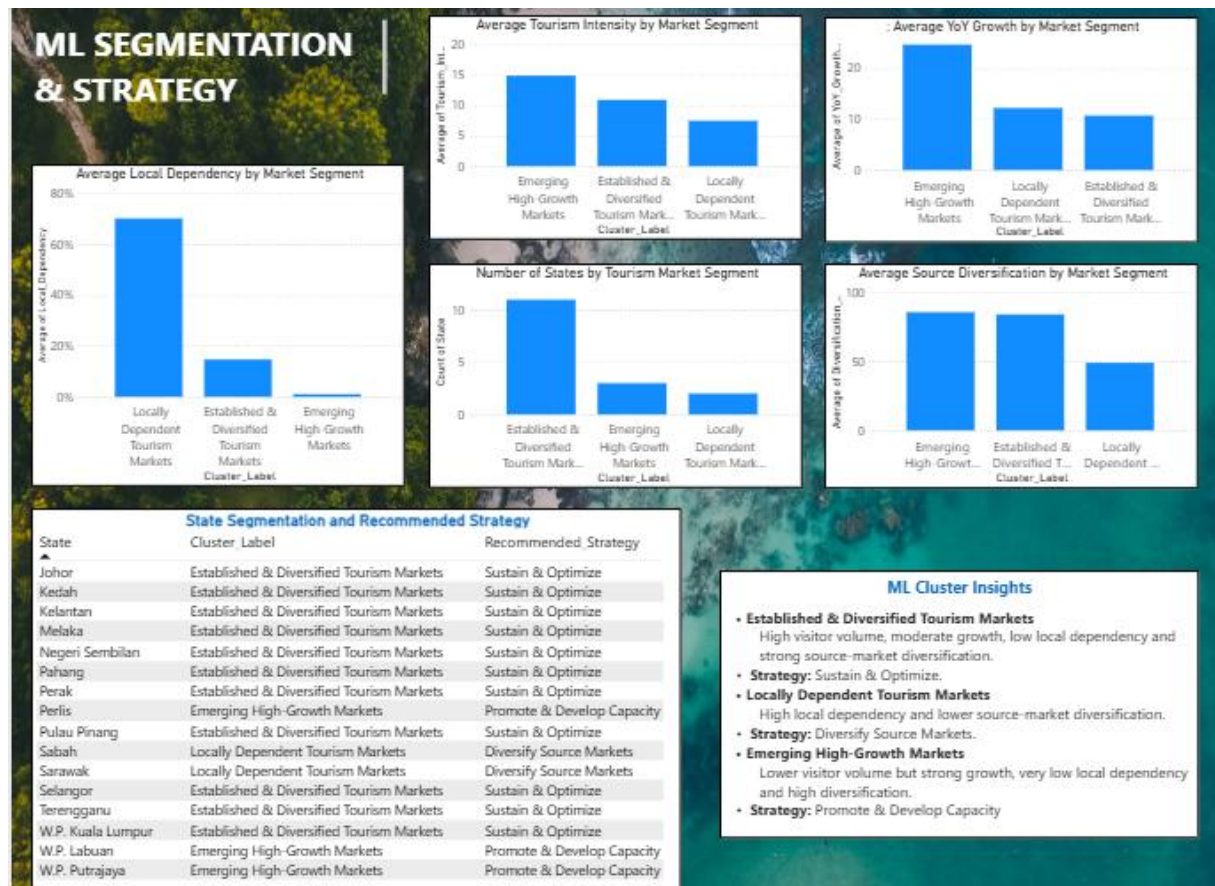


Figure 3. ML Segmentation & Strategy dashboard showing tourism market segments, cluster profiles and recommended strategies.

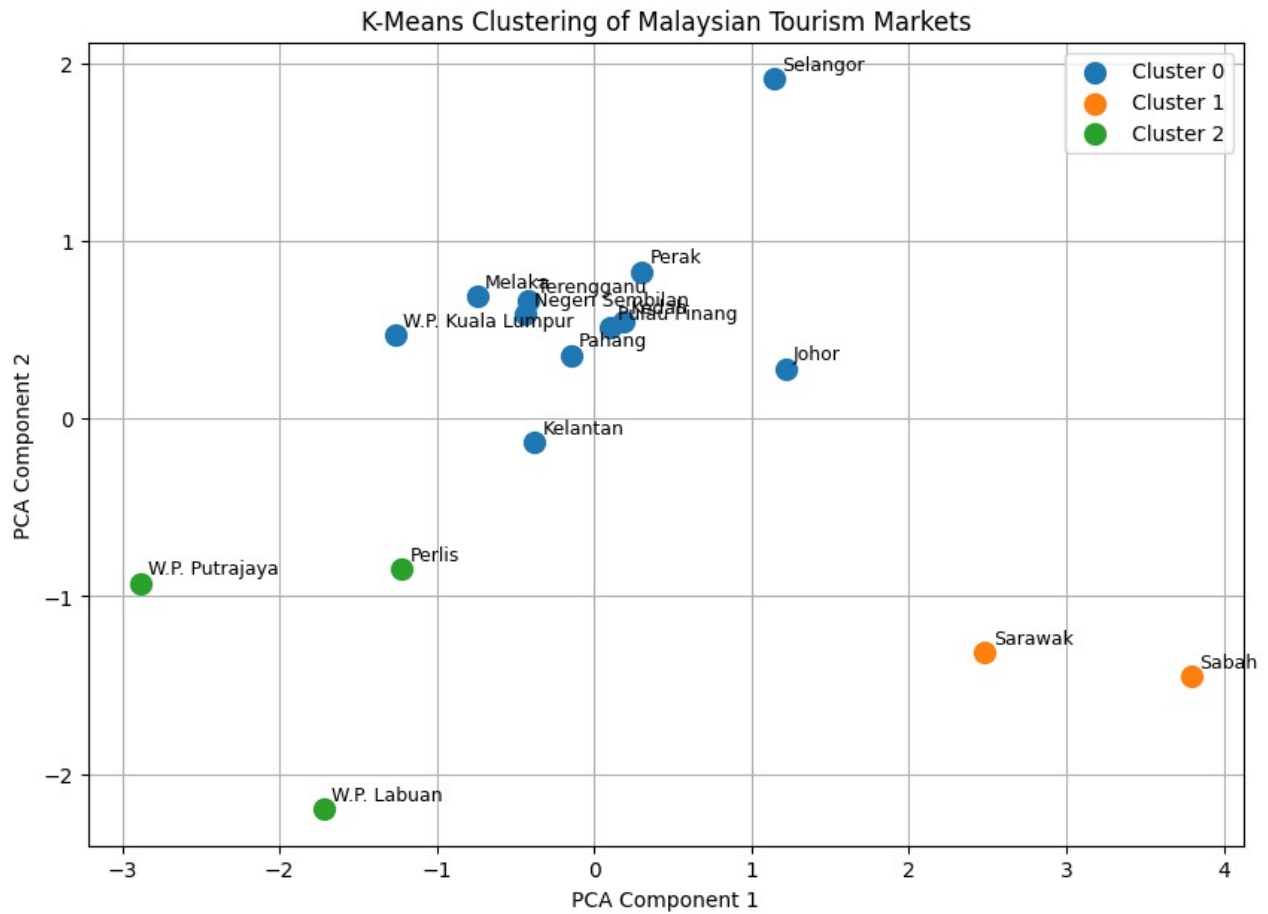


Figure 4. PCA visualisation of the K-Means state clusters (visualisation only; clustering was performed on the five standardised features).

CONCLUSION AND RECOMMENDATIONS

The analysis demonstrates that Malaysia's domestic tourism market cannot be understood through visitor totals alone. The 2025 market combines strong national recovery with significant differences in state-level volume, growth, local dependency, source diversification and tourism intensity. These indicators provide a richer basis for state-level planning than a uniform national strategy.

The clustering results translate these differences into three actionable tourism-market segments. The following strategies are recommended:

- **Established & Diversified Tourism Markets:** sustain demand, optimise visitor yield and length of stay, manage capacity at high-demand destinations and use targeted campaigns to protect market diversity.
- **Locally Dependent Tourism Markets:** diversify origin markets through interstate promotion, thematic travel packages and improved connectivity; monitor over-reliance on local demand as a resilience risk.
- **Emerging High-Growth Markets:** promote market visibility while developing tourism capacity, services and destination readiness so that rapid growth can be accommodated without reducing visitor experience.

Overall, the project contributes a data-driven framework that converts descriptive statistics into market-structure indicators, machine-learning segments and state-specific recommendations. This approach can support more targeted and resilient domestic tourism planning in Malaysia.

LIMITATIONS

- The tourism-flow and clustering analysis is based on a 2025 snapshot and does not capture multi-year changes in origin-destination relationships.
- Domestic visitor statistics represent visitor movements and may include repeat trips; they should not be interpreted as unique individuals.
- The K-Means model contains only 16 state/Federal Territory observations, so cluster results should be interpreted as exploratory segmentation rather than a definitive classification.
- The analysis supports tourism planning but does not directly measure environmental, social or ecological sustainability outcomes.
- Cluster labels and strategies are analytical interpretations based on the observed profiles; they are not official DOSM classifications.

REFERENCES

- Department of Statistics Malaysia (DOSM). (2025). Domestic Tourism Survey, Malaysia. Putrajaya: DOSM. <https://open.dosm.gov.my/publications>
- Department of Statistics Malaysia (DOSM). (2026). Population by State [Open data dataset]. data.gov.my. https://data.gov.my/data-catalogue/population_state
- MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, 1, 281–297.
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53–65.
- United Nations. Sustainable Development Goal 8, Target 8.9; Sustainable Development Goal 12, Target 12.b.